Claude Code Workshop

Introduction

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Resources

- https://dunnlab.org/dunnlab_code/
- https://github.com/caseywdunn/dunnlab_code
- Ask claude code about itself

What are these things?

- Artificial Intelligence (AI)
  - Neural networks
    - Large Language Models (LLMs)
      - Transformers
        - Claude (anthropic)
        - ChatGPT (openai)
        - Gemini (google)

What are these things?

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

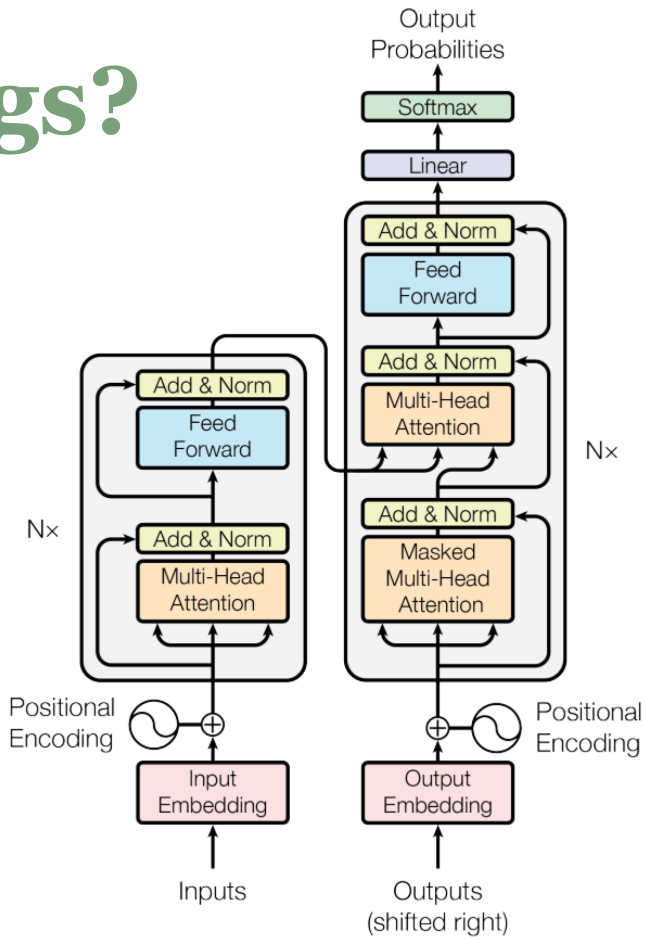


Figure 1: The Transformer - model architecture.

What are these things?

A curriculum to understand transformers

- https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi
- https://www.youtube.com/watch?v=VMj-3S1tku0
- https://karpathy.github.io/2026/02/12/microgpt/

Chatbots vs. Agents

You ask questions and make requests of **chatbots**. They respond with text, and you can also upload and download files.

You ask questions and make requests of **agents**. They then act on your behalf. This can be on your computer or in the world at large.

Chatbots vs. Agents

Comparison of debugging workflows with chatbots vs. agents...

Claude

Anthropic’s flagship model and the products built around them.

Opus – their most powerful model

Sonnet – a smaller cheaper model that is still plenty good for most tasks

Using Claude

Claude is bundled into several products.

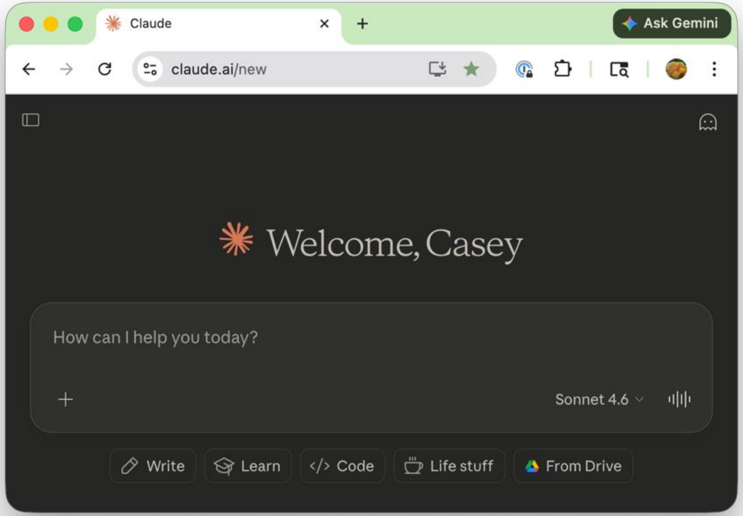
Chat – a chatbot

Cowork – a general-purpose assistant agent

Code – a coding agent

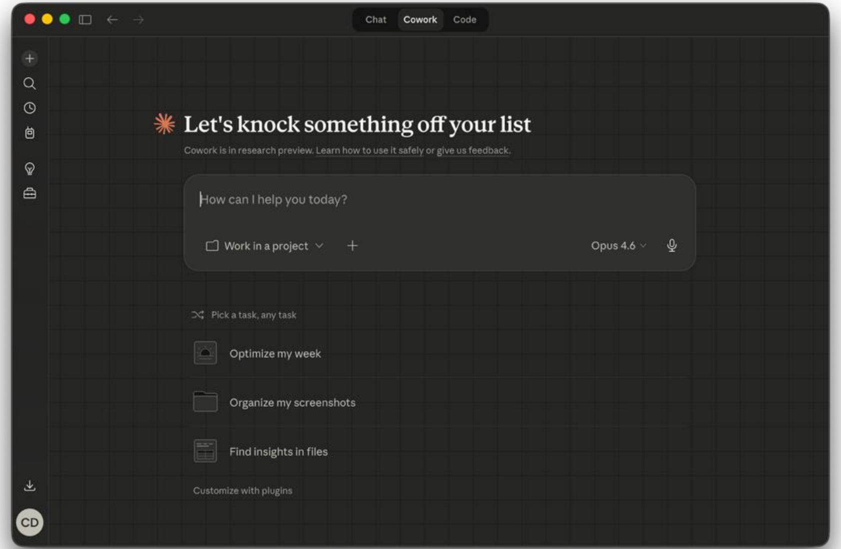
Claude Chat

Chat is familiar to anyone who has used any chatbot.



Claude Cowork

Is an agent built on Claude code but intended for non-technical users. Scheduling, batch processing documents, etc…



Claude plans

To use Claude Code and Cowork you need a paid plan

Claude plans

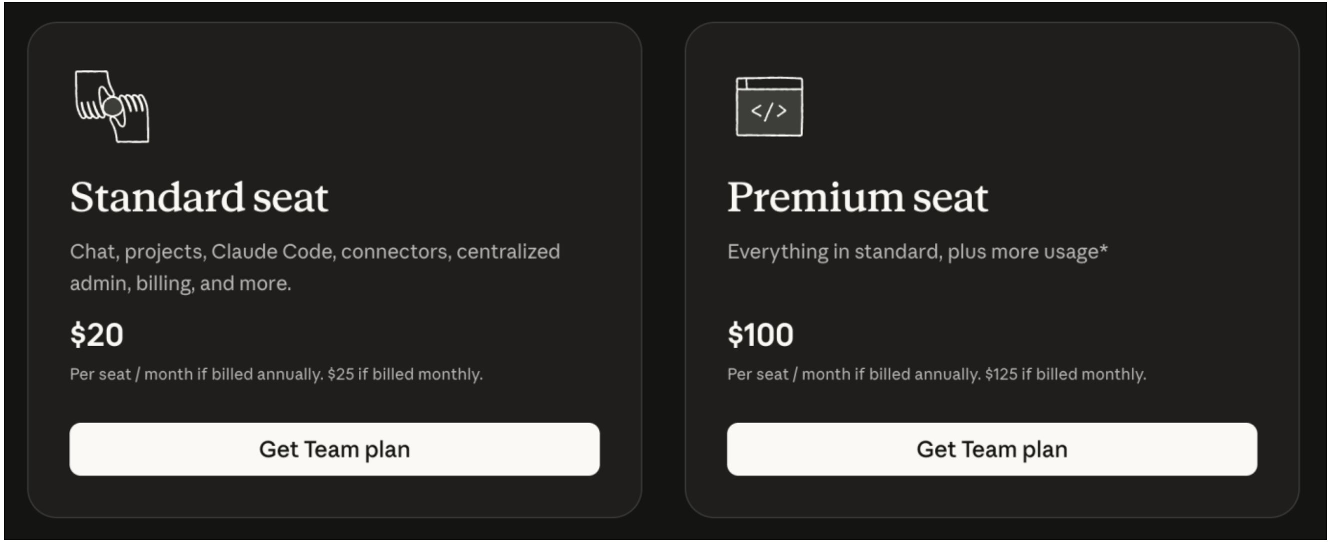
Pro plan is generally the place to start. Can bump to Max later if needed.

Plan	Price	Billing Interval	Usage Capacity	Best For
Free	$0	N/A	Limited	Occasional use
Pro	$20/month $200/year	Monthly or annual	Standard	Regular use
Max 5x	$100	Monthly	5x Pro capacity per session	Frequent users who work with Claude on a variety of tasks
Max 20x	$200	Monthly	20x Pro capacity per session	Daily users who collaborate often with Claude for most tasks

https://support.claude.com/en/articles/11049762-choosing-a-claude-plan

Claude plans

The team plan simplifies billing if you are purchasing for 5 or more people (eg a whole lab).



https://claude.com/pricing/team

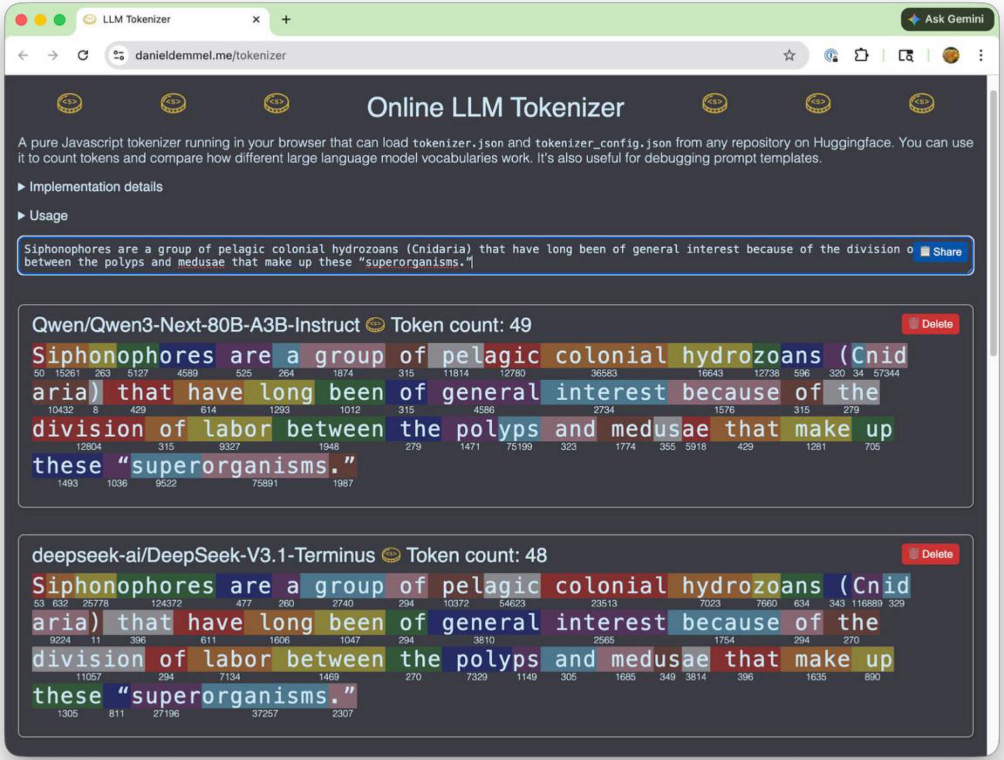
Tokens

Tokens can mean two very different things:

- A cryptographic key used to authenticate a service
- The lexical unit that LLMs operate with

Tokens

Each token is about 0.75 words



Tokens

Different plans have different usage caps, where usage is measured by tokens weighted by:

- Which model you use (eg Opus vs Sonnet)
- Whether it is input or output (output costs more)

Tokens

Claude plans have 5 hour limits and weekly limits

When you hit your limit, you can:

- Wait until it resets (touch grass)
- Buy more tokens (pay as you go)
- Upgrade your plan

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Security

Security

The CIA triad of computer security threats:

Confidentiality — Stealing or exposing secrets. Unauthorized parties gain access to sensitive data (e.g., data breaches, eavesdropping, credential theft).

Integrity — Modifying things. Data or systems are altered without authorization (e.g., tampering with records, man-in-the-middle attacks, malware that corrupts files).

Availability — Breaking things. Systems or data become inaccessible to legitimate users (e.g., ransomware, DDoS attacks, hardware destruction).

Security

Giving a new, alien robot agent access to your computer opens you (and others) up to all three categories of threats!

It is doing things on your computer.

It is sending your data to Anthropic's servers

It may send and retrieve data from other servers

Security

Specific scenarios:

- You violate HIPAA
- Secure information (passwords, crypto wallets, financial data, compromising personal information) are leaked
- A malicious actor takes control of the agent (eg prompt injection)
- Malicious code is injected into your project that you spread to others

Security

There is always a tradeoff between

- Giving the agent too much free rein and it does something bad
- Highly restricting the agent and it is cumbersome to do anything good

Security

System permissions

Hard limits set by computer on what the agent can do

- Read/ write permissions
- Execute permissions
- Firewalls

Agent permissions

Ask the agent nicely to follow your suggestions:

- Allow
- Ask
- Deny

Security

General approach to security:

- Control everything you can at system level
- Set agent permissions to be more restrictive where needed

System level permissions

There are a few ways to restrict system level permissions

- Use a dedicated computer with no private data
- Use a virtual machine/ container (eg Docker)
- Have a dedicated user account for agents

System level permissions

My own use cases for claude code include:

- My own user account with moderate agent permissions
- A dedicated user account or container few or no agent permission limits
- On a YCRC cluster with extremely strict permissions

System level permissions

Running claude with

```
claude --dangerously-skip-permissions
```

Turns off claude permissions systems.

It can code for hours without oversight.

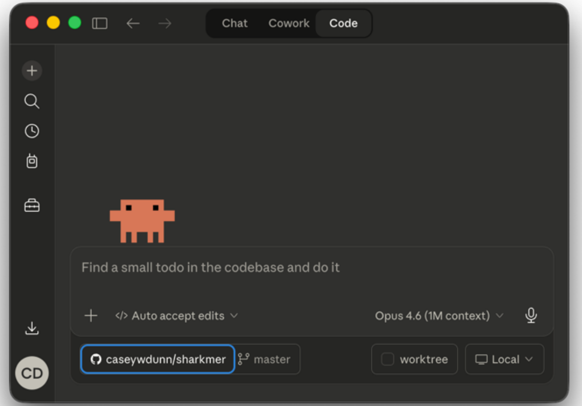
Only do this with strong system security in place.

Claude code is part of an ecosystem

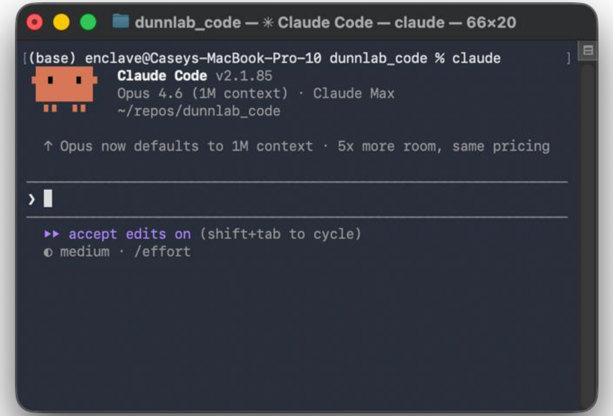
In our lab:

- **VS Code** – Interactive Development Environment (IDE), a glorified text editor
- **Git and GitHub** – Version control
- **Python** for most things, **R** when we need a specific library, **Rust** when we need high performance code
- **conda** and docker for analysis environments
- **YCRC** Clusters
- **claude code**

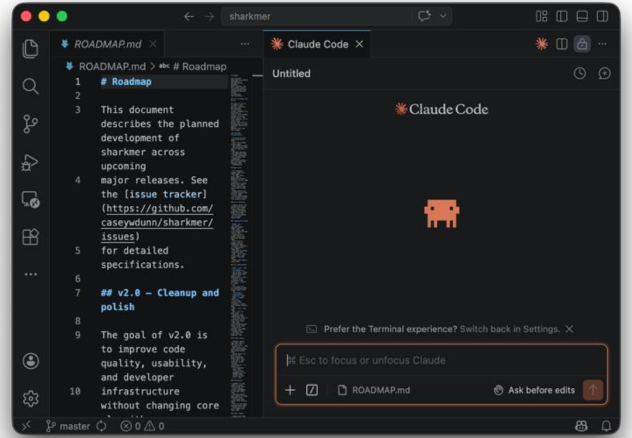
Using claude code



Desktop Application



Command Line Interface



VS Code Extension

Using claude code

You specify a working directory – this is the set of files claude will focus on

But expect that it can access any files without explicit permissions restrictions

Using claude code

You can just open claude code and start getting great results, especially for small and simple projects.

Using claude code

But you need to learn how to use the tool to get good results on large or complex projects.

The two main things to know:

1. How to manage the model **context window**
2. How to make a **useful plan**

Model Context Window

Think of it as the working memory, the RAM, of the model instance.

It is measured in tokens.

2,048	ChatGPT 3
128,000	ChatGPT 5
1,000,000	Opus 4.7

Model Context Window

Even with a very large context window, you need to manage it for it to work well.

Context will

- Fill up. You will run out of space.
- Rot. When it has been used for a while its information content degrades.

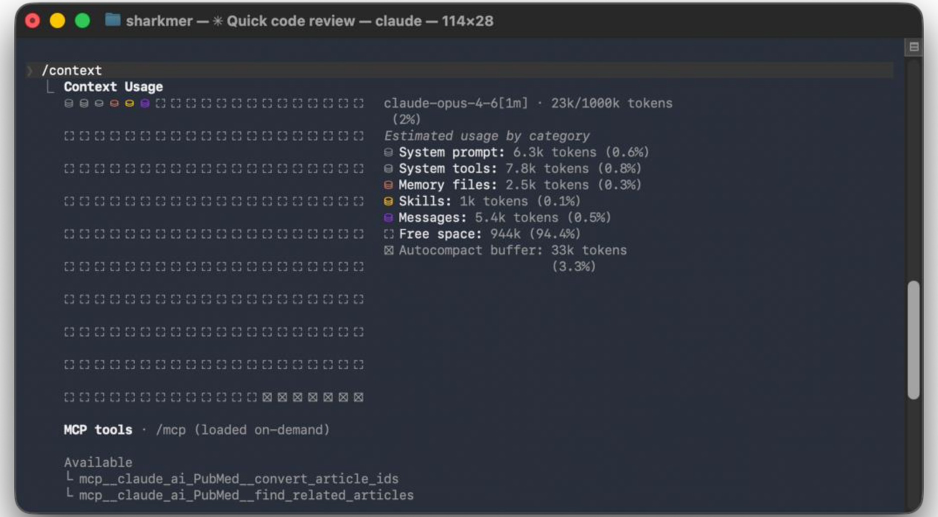
Model Context Window

Managing Context

- Clearing – clear the context between tasks
- Gatekeeping – don't bring things into context if they aren't needed

Model Context Window

`/context` to view context usage



`/clear` to clear context

Gatekeeping context

`CLAUDE.md` is a special file you can put in the root of your project folder. It is the first thing claude code reads when it starts up.

Keep it short – under 100 lines.

Explain what the project is, big picture approach, any big dos or don’ts, and tell it what info is in other files.

Don’t write it yourself, prompt claude code to write it

Gatekeeping context

`CLAUDE.md` is a special file you can put in the root of your project folder. It is the first thing claude code reads when it starts up. Project specific information.

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Explain what the project is, big picture approach, any big dos or don'ts, and tell it what info is in other files.

Don't write it yourself, prompt claude code to write it

Gatekeeping context

Skills are special structured files.

They contain a very short description and detailed instructions.

The descriptions of all installed skills are loaded into context. The model decides when to load the full instructions into context based on the description.

Great for repeated tasks.

Planning

First prompt claude to develop a plan. Ask it to put that plan in the CLAUDE.md and linked markdown files.

Work with it iteratively to refine the plan.

Then ask it to make a checklist for the plan.

Simple development loop

In a single prompt, ask claude to:

1. Tackle the next incomplete task
2. Check off the task when done
3. Commit changes (if using git)

Then clear context and repeat.

General advice

1. Think of claude as its own manual. Ask it how to use it.
2. Think big. When you start using it, you will be scoping tasks based on them being harder than they are now. Ask it to do bigger things than you would do manually.
3. It isn't just for coding – it can tell you about methods, suggest next steps, help interpret science.
4. Be skeptical. You are a scholar, Claude is a tool.

Thanks.